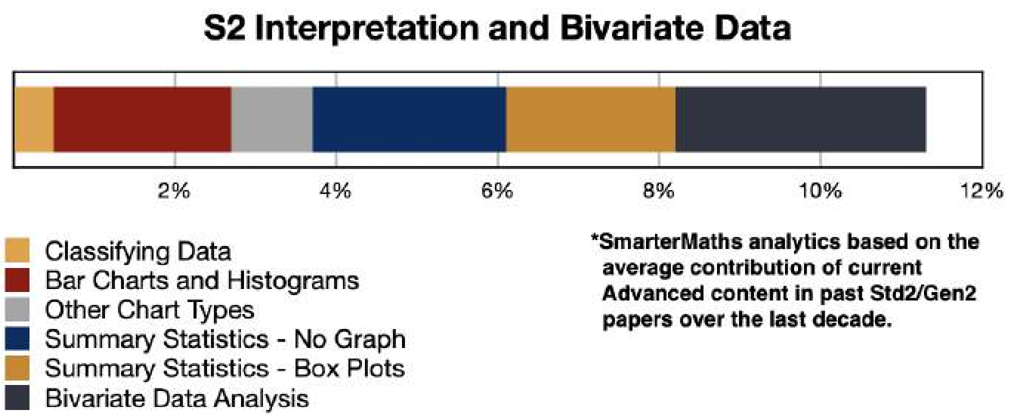
ADV: Statistics (Adv), S2 Interpretation and Bivariate Data (Adv)

Bivariate Data Analysis (Y12)

Teacher: Ljiljana Todic

Exam Equivalent Time: 93 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*



HISTORICAL CONTRIBUTION

- *S2 Interpretation and Bivariate Data* is a Year 12 topic that hasn't previously existed in the Advanced course, although it has a decade long history in the Std2/Gen2 exam.
- It provides an area where examiners can test common content between the *Advanced* and *Standard 2* courses. The above bar chart shows the relative importance of the subtopics in past Std2/Gen2 exams.
- *S2 Interpretation and Bivariate Data* has been split into six sub-topics for analysis which are listed in the bar chart above.
- This analysis looks at *Bivariate Data Analysis (3.1%).*

HSC ANALYSIS - What to expect and common pitfalls

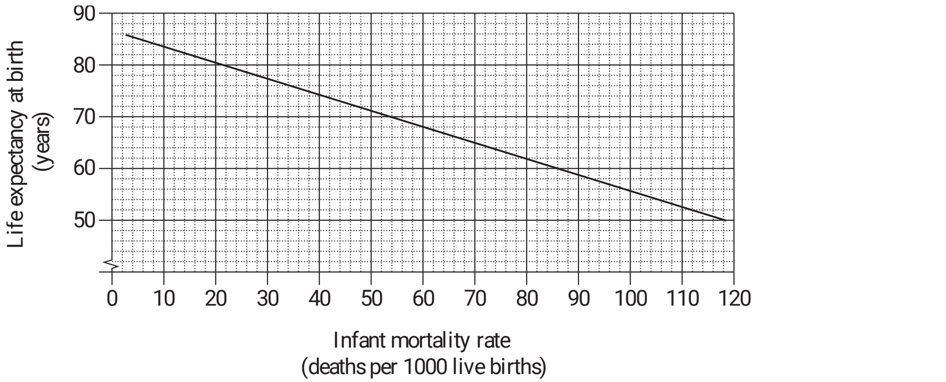
- *Bivariate Data Analysis (3.1%)* investigates scatterplots, correlation, lines of best fit, etc ...
- The *2020 Advanced exam* allocated a significant 5 marks to this topic, and even more importantly, presented a word-heavy narrative style that is critical revision.
- Past questions have proven challenging, with every longer answer question asked in the last 10 years producing a sub-50% mean mark in at least one part.
- Calculating a least squares regression line from raw data is a key skill that students must be able to execute efficiently. *EQ-Bank Q1-3* are important revision examples covering this area.
- ***Pitfalls:*** Marker's comments have highlighted issues in finding equations of best fit, interpreting gradients and identifying limitations of an equation - all areas well covered in this question database.

Questions

1. Algebra, STD2 A2 2017 HSC 3 MC

RAP Data - Bottom 22%: School result (93%) was 6% above state average (87%)

The graph shows the relationship between infant mortality rate (deaths per 1000 live births) and life expectancy at birth (in years) for different countries.



What is the life expectancy at birth in a country which has an infant mortality rate of 60?

A. 68 years

B. 69 years

C. 86 years

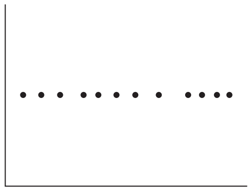
D. 88 years

2. Statistics, STD2 S4 2017 HSC 12 MC

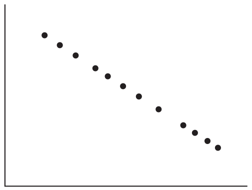
RAP Data - Bottom 24%: School result (85%) was 7% above state average (78%)

Which of the data sets graphed below has the largest positive correlation coefficient value?

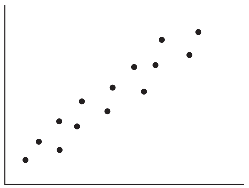
A.



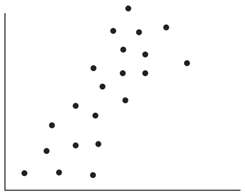
B.



C.



D.



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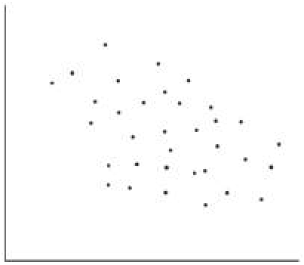
3. Statistics, STD2 S4 2016 HSC 3 MC

The graph shows a scatterplot for a set of data.



Which of the following is the best approximation for the correlation coefficient of this set of data?

(A) -1

(B) -0.3

(C) 0.3

(D) 1

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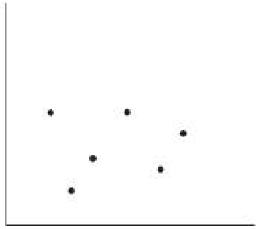
4. Statistics, STD2 S4 2011 HSC 8 MC

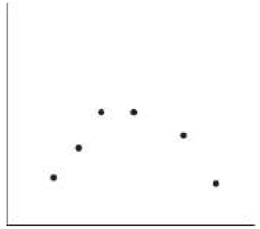
RAP Data - Bottom 10%: School result (70%) was 1% above state average (69%)

In which graph would the data have a correlation coefficient closest to -0.9?

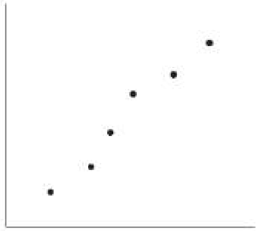
(A)



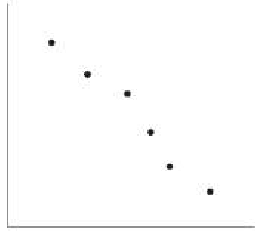
(B)



(C)



(D)



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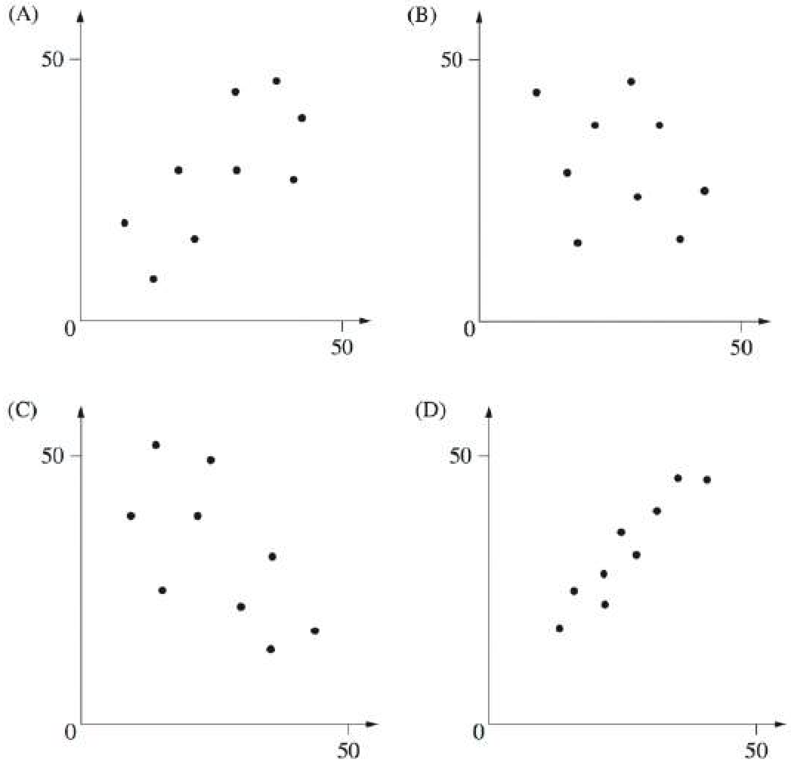
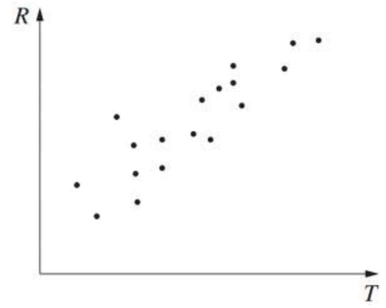
5. Statistics, STD2 S4 2013 HSC 2 MC

Which graph best shows data with a correlation closest to 0.3?



6. Statistics, STD2 S4 2008 HSC 12 MC

A scatterplot is shown.



Which of the following best describes the correlation between R and T?

(A) Positive

(B) Negative

(C) Positively skewed

(D) Negatively skewed

7. Statistics, STD2 S4 2007 HSC 9 MC

Which of the following would be most likely to have a positive correlation?

(A) The population of a town and the number of schools in that town

(B) The price of petrol per litre and the number of litres of petrol sold

(C) The hours training for a marathon and the time taken to complete the marathon

(D) The number of dogs per household and the number of televisions per household

8. Statistics, STD2 S4 2012 HSC 11 MC

RAP Data - Bottom 5%: School result (41%) was -2% below state average (43%)

Which of the following relationships would most likely show a negative correlation?

(A) The population of a town and the number of hospitals in that town.

(B) The hours spent training for a race and the time taken to complete the race.

(C) The price per litre of petrol and the number of people riding bicycles to work.

(D) The number of pets per household and the number of computers per household.

9. Statistics, STD2 S4 SM-Bank 1

A student claimed that as time spent swimming training increases, the time to run a 1 kilometre time trial decreases.

After collecting and analysing some data, the student found the correlation coefficient, r, to be –0.73.

What does this correlation indicate about the relationship between the time a student spends swimming training and their 1 kilometre run time trial times. ***(1 mark)***

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10. Statistics, STD2 S4 EQ-Bank 2

Pedro is planning a statistical investigation.

List the steps that Pedro must follow to execute the statistical investigation correctly. ***(2 marks)***

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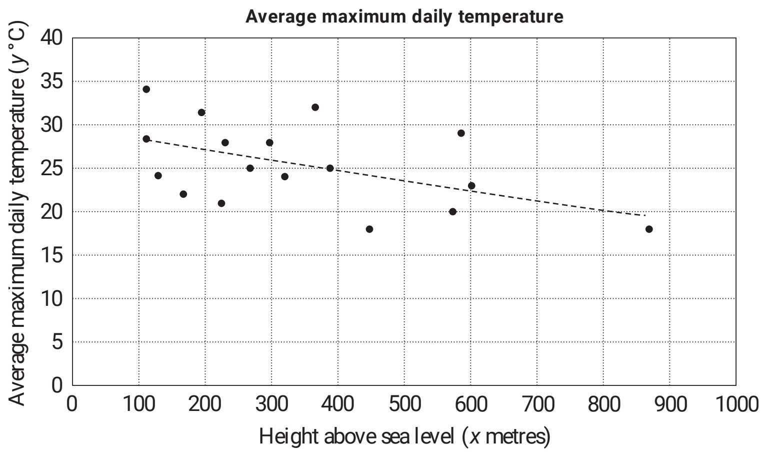
11. Statistics, 2ADV S2 2021 HSC 17

For a sample of 17 inland towns in Australia, the height above sea level, x (metres), and the average maximum daily temperature, y (°C), were recorded.

The graph shows the data as well as a regression line.



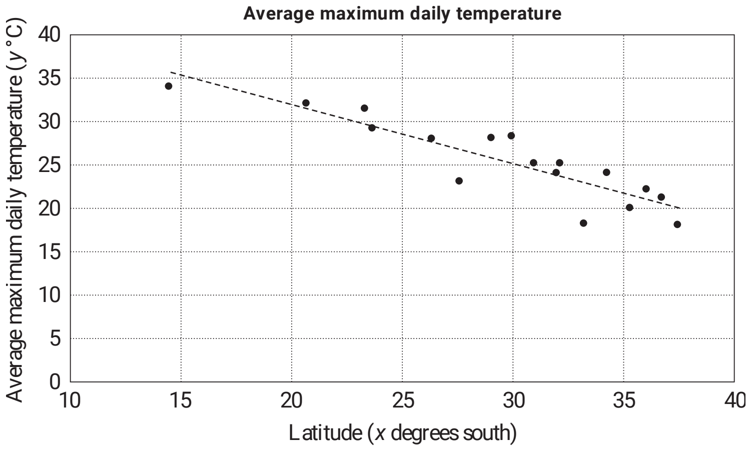
The equation of the regression line is $y = 29.2 - 0.011x$.

The correlation coefficient is $r = -0.494$.

a. **i.** By using the equation of the regression line, predict the average maximum daily temperature, in degrees Celsius, for a town that is 540 m above sea level. Give your answer correct to one decimal place. ***(1 mark)***

ii. The gradient of the regression line is −0.011. Interpret the value of this gradient in the given context. ***(2 marks)***

b. The graph below shows the relationship between the latitude, x (degrees south), and the average maximum daily temperature, y (°C), for the same 17 towns, as well as a regression line.



The equation of the regression line is $y = 45.6 - 0.683x$.

The correlation coefficient is $r = -0.897$.

Another inland town in Australia is 540 m above sea level. Its latitude is 28 degrees south.

Which measurement, height above sea level or latitude, would be better to use to predict this town's average maximum daily temperature? Give a reason for your answer. ***(1 mark)***

12. Statistics, 2ADV S2 EQ-Bank 2

The table below lists the average *body weight* (in kilograms) and average *brain weight* (in grams) of nine animal species.

species	body weight (kg)	brain weight (g)
baboon	10.55	179.5
cat	3.30	25.6
goat	27.70	115.0
guinea pig	1.04	5.5
rabbit	2.50	12.1
rat	0.28	1.9
red fox	4.24	50.4
rhesus monkey	6.80	179.0
sheep	55.50	175.0

A least squares regression line is fitted to the data using *body weight* as the independent variable.

i. Calculate the equation of the least squares regression line. ***(1 mark)***

ii. If dingos have an average body weight of 22.3 kilograms, calculate the predicted average brain weight of a dingo using your answer to part i. ***(1 mark)***

13. Statistics, 2ADV S2 EQ-Bank 3

The table below lists the average *life span* (in years) and average *sleeping time* (in hours/day) of 9 animal species.

species	life span (years)	sleeping time (hours/day)
baboon	27	10
cow	30	4
goat	20	4
guinea pig	8	8
horse	46	3
mouse	3	13
pig	27	8
rabbit	18	8
rat	5	13

i. Using *sleeping time* as the independent variable, calculate the least squares regression line. ***(1 mark)***

ii. A wallaby species sleeps for 4.5 hours, on average, each day.

Use your equation from **part i** to predict its expected *life span*, to the nearest year. ***(1 mark)***

14. Statistics, 2ADV S2 EQ-Bank 1

The arm spans (in cm) and heights (in cm) for a group of 13 boys have been measured. The results are displayed in the table below.

Arm span (cm)	Height (cm)
152	152
153	155
174	168
141	149
170	172
165	168
163	163
155	157
165	165
152	150
143	146
156	153
174	174

The aim is to find a linear equation that allows arm span to be predicted from height.

a. What will be the independent variable in the equation? ***(1 mark)***

b. Assuming a linear association, determine the equation of the least squares regression line that enables *arm span* to be predicted from *height*. Write this equation in terms of the variables *arm span* and *height*. Give the coefficients correct to two decimal places. ***(2 marks)***

c. Using the equation that you have determined in **part b.**, interpret the slope of the least squares regression line in terms of the variables *height* and *arm span*. ***(1 mark)***

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15. Statistics, STD2 S4 EQ-Bank 4

Ten high school students have their height and the length of their right foot measured.

The results are recorded in the table below.

Height (cm)	142	164	183	172	154	158	149	181	175	177
Foot (cm)	20	26	28	26	22	23	24	29	27	27

i. Using technology, calculate Pearson's correlation coefficient for the data. Give your answer to 3 decimal places. ***(1 mark)***

ii. Describe the strength of the association between height and length of right foot for these students. ***(1 mark)***

iii. Using technology, determine the least squares regression line that allows height to be predicted from right foot length. ***(1 mark)***

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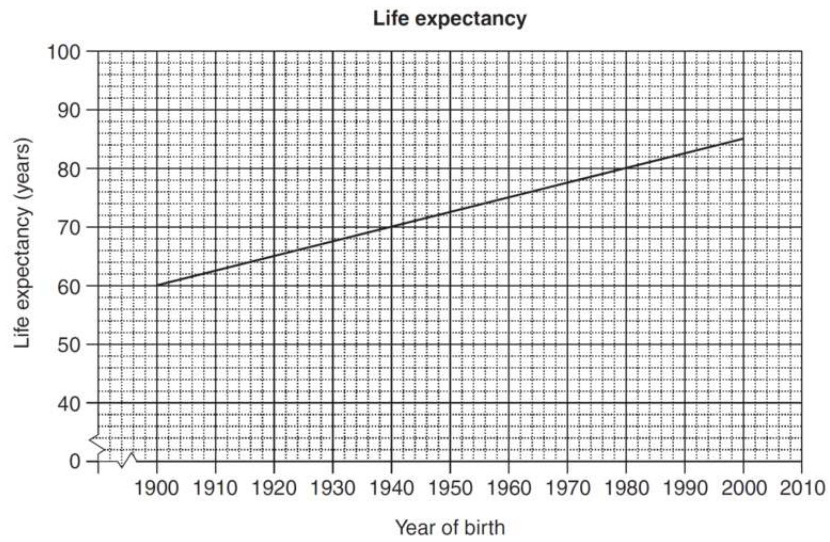
16. Algebra, STD2 A2 2016 HSC 29e

Part i: RAP Data - Bottom 17%: School result (87%) was 4% above state average (83%)

The graph shows the life expectancy of people born between 1900 and 2000.



i. According to the graph, what is the life expectancy of a person born in 1932? ***(1 mark)***

ii. With reference to the value of the gradient, explain the meaning of the gradient in this context. ***(2 marks)***

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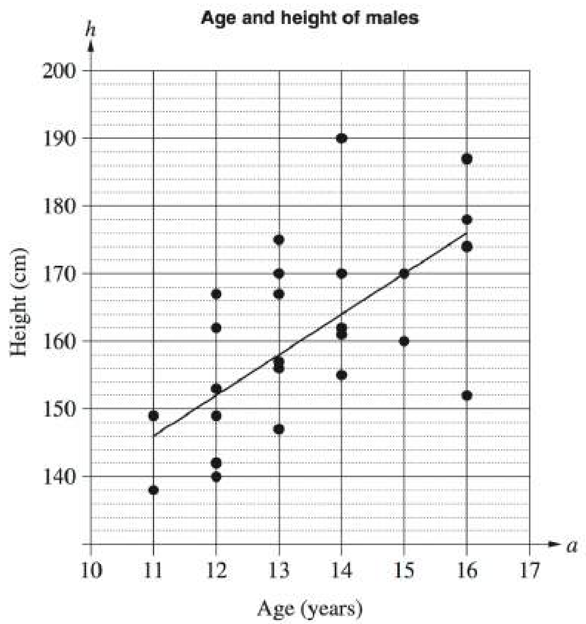
17. Statistics, STD2 S4 2013 HSC 28b

Part v: RAP Data - Bottom 19%: School result (67%) was 5% above state average (62%)

Ahmed collected data on the age (a) and height (h) of males aged 11 to 16 years.

He created a scatterplot of the data and constructed a line of best fit to model the relationship between the age and height of males.



i. Determine the gradient of the line of best fit shown on the graph. ***(1 mark)***

ii. Explain the meaning of the gradient in the context of the data. ***(1 mark)***

iii. Determine the equation of the line of best fit shown on the graph. ***(2 marks)***

iv. Use the line of best fit to predict the height of a typical 17-year-old male. ***(1 mark)***

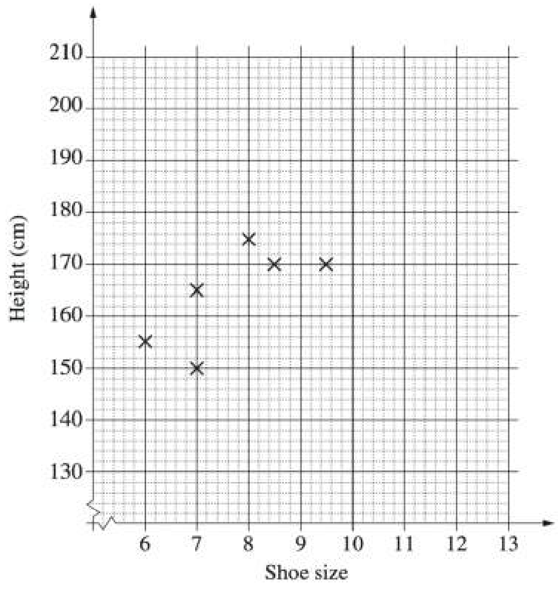
v. Why would this model not be useful for predicting the height of a typical 45-year-old male? ***(1 mark)***

18. Statistics, STD2 S4 2015 HSC 28e

The shoe size and height of ten students were recorded.

Shoe size	6	7	7	8	8.5	9.5	10	11	12	12
Height	155	150	165	175	170	170	190	185	200	195

i. Complete the scatter plot AND draw a line of fit by eye. ***(2 marks)***

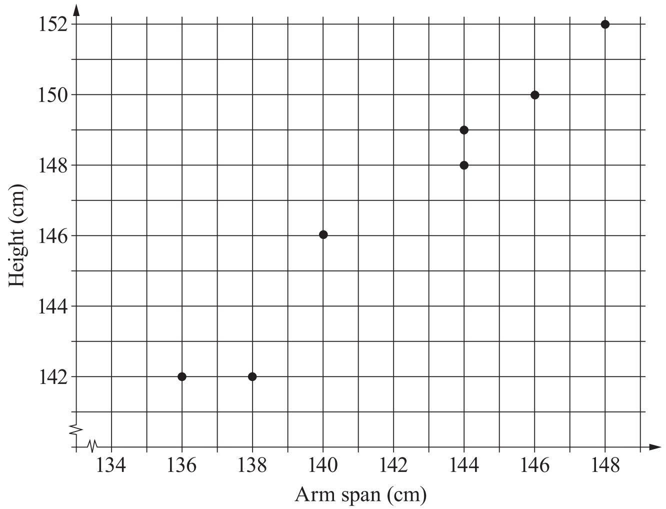


ii. Use the line of fit to estimate the height difference between a student who wears a size 7.5 shoe and one who wears a size 9 shoe. ***(1 mark)***

iii. A student calculated the correlation coefficient to be 1 for this set of data. Explain why this cannot be correct. ***(1 mark)***

19. Statistics, STD2 S4 2019 HSC 23

A set of bivariate data is collected by measuring the height and arm span of seven children. The graph shows a scatterplot of these measurements.



a. Calculate Pearson's correlation coefficient for the data, correct to two decimal places. ***(1 mark)***

b. Identify the direction and the strength of the linear association between height and arm span. ***(1 mark)***

c. The equation of the least-squares regression line is shown.

Height = 0.866 × (arm span) + 23.7

A child has an arm span of 143 cm.

Calculate the predicted height for this child using the equation of the least-squares regression line. ***(1 mark)***

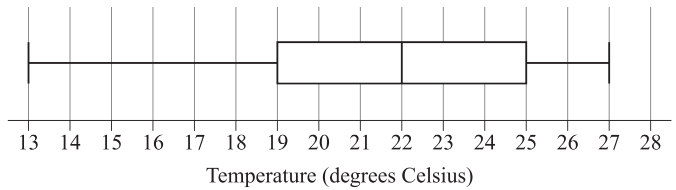
20. Statistics, 2ADV S2 2020 HSC 27

A cricket is an insect. The male cricket produces a chirping sound.

A scientist wants to explore the relationship between the temperature in degrees Celsius and the number of cricket chirps heard in a 15-second time interval.

Once a day for 20 days, the scientist collects data. Based on the 20 data points, the scientist provides the information below.

- A box-plot of the temperature data is shown.



- The mean temperature in the dataset is 0.525°C below the median temperature in the dataset.
- A total of 684 chirps was counted when collecting the 20 data points.

The scientist fits a least-squares regression line using the data (x, y), where x is the temperature in degrees Celsius and y is the number of chirps heard in a 15-second time interval. The equation of this line is

$$y = -10.6063 + bx,$$

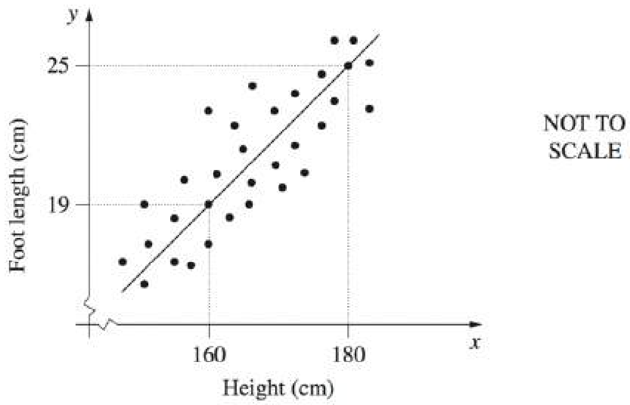
where b is the slope of the regression,

The least-squares regression line passes through the point $(\overline{x}, \overline{y})$, where $\overline{x}$ is the sample mean of the temperature data and $\overline{y}$ is the sample mean of the chirp data.

Calculate the number of chirps expected in a 15-second interval when the temperature is 19° Celsius. Give your answer correct to the nearest whole number. ***(5 marks)***

21. Statistics, STD2 S4 2006 HSC 27b

Each member of a group of males had his height and foot length measured and recorded. The results were graphed and a line of fit drawn.



i. Why does the value of the y-intercept have no meaning in this situation? ***(1 mark)***

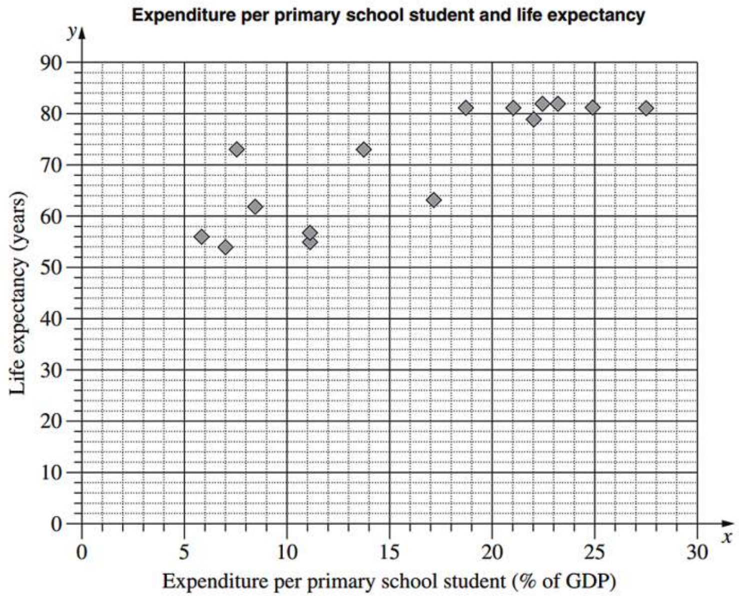
ii. George is 10 cm taller than his brother Harry. Use the line of fit to estimate the difference in their foot lengths. ***(1 mark)***

iii. Sam calculated a correlation coefficient of -1.2 for the data. Give TWO reasons why Sam must be incorrect. ***(2 marks)***

22. Statistics, STD2 S4 2014* HSC 30b

Part viii: RAP Data - Bottom 10%: School result (2%) was 2% above state average (0%)

The scatterplot shows the relationship between expenditure per primary school student, as a percentage of a country's Gross Domestic Product (GDP), and the life expectancy in years for 15 countries.



i. For the given data, the correlation coefficient, r, is 0.83. What does this indicate about the relationship between expenditure per primary school student and life expectancy for the 15 countries? ***(1 mark)***

ii. For the data representing expenditure per primary school student, Q_L is 8.4 and Q_U is 22.5. What is the interquartile range? ***(1 mark)***

iii. Another country has an expenditure per primary school student of 47.6% of its GDP. Would this country be an outlier for this set of data? Justify your answer with calculations. ***(2 marks)***

iv. On the scatterplot, draw the least-squares line of best fit $y = 1.29x + 49.9$. ***(2 marks)***

v. Using this line, or otherwise, estimate the life expectancy in a country which has an expenditure per primary school student of 18% of its GDP. ***(1 mark)***

vi. Why is this line NOT useful for predicting life expectancy in a country which has expenditure per primary school student of 60% of its GDP? ***(1 mark)***

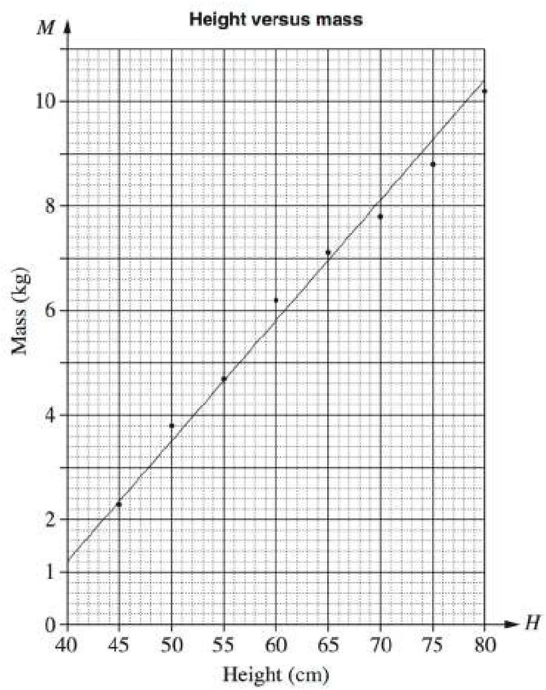
23. Statistics, STD2 S4 2009 HSC 28b

Part i: RAP Data - Bottom 13%: School result (51%) was 3% above state average (48%)

The height and mass of a child are measured and recorded over its first two years.

Height (cm), H	45	50	55	60	65	70	75	80
Mass (kg), M	2.3	3.8	4.7	6.2	7.1	7.8	8.8	10.2

This information is displayed in a scatter graph.



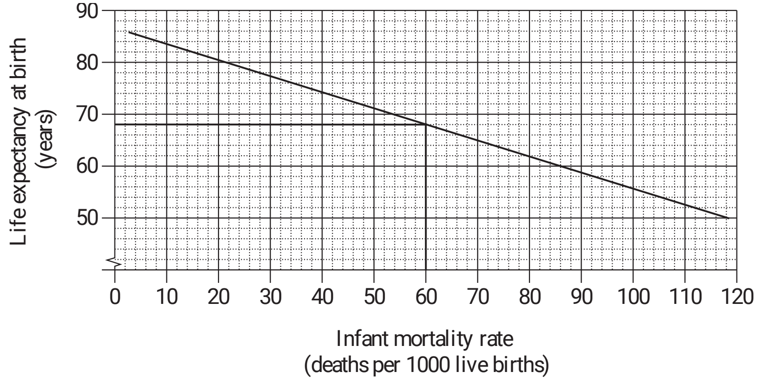
i. Describe the correlation between the height and mass of this child, as shown in the graph. ***(1 mark)***

ii. A line of best fit has been drawn on the graph.
Find the equation of this line. ***(2 marks)***

Worked Solutions

1. Algebra, STD2 A2 2017 HSC 3 MC

When infant mortality rate is 60, life expectancy at birth is 68 years (see below).



$\Rightarrow A$

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2. Statistics, STD2 S4 2017 HSC 12 MC

Largest positive correlation occurs when both variables move in tandem. The tighter the linear relationship, the higher the correlation.

$\Rightarrow C$

(Note that B is negatively correlated)

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3. Statistics, STD2 S4 2016 HSC 3 MC

Correlation is negative and weak.

$\Rightarrow B$

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4. Statistics, STD2 S4 2011 HSC 8 MC

Data needs to show a strong negative correlation (i.e. top left to lower right.)

$\Rightarrow D$

Worked Solutions

5. Statistics, STD2 S4 2013 HSC 2 MC

A is correct since the data slopes bottom left to top right (i.e. it's positive).

D also slopes correctly but exhibits a higher correlation co-efficient.

$\Rightarrow A$

6. Statistics, STD2 S4 2008 HSC 12 MC

Correlation is positive.

NB. The skew does not refer to correlation.

$\Rightarrow\ A$

7. Statistics, STD2 S4 2007 HSC 9 MC

Positive correlation means that as one variable increases, the other tends to increase also.

$\Rightarrow\ A$

8. Statistics, STD2 S4 2012 HSC 11 MC

Increased hours training should reduce the time to complete a race.

$\Rightarrow\ B$

♦ Mean mark 43%

9. Statistics, STD2 S4 SM-Bank 1

– 0.73 indicates a strong negative relationship exists.

In this case, it means the more time spent swimming training is associated with a quicker time of running a 1 kilometre time trial.

10. Statistics, STD2 S4 EQ-Bank 2

Steps are:

- Identify a problem and pose a statistical question
- Collect or obtain data
- Represent and analyse the data collected or obtained
- Communicate and interpret findings.

11. Statistics, 2ADV S2 2021 HSC 17

a.i. $y = 29.2 - 0.011(540)$

$= 23.26$

$= 23.3^\circ\text{C}\ \ (1\text{ d.p.})$

a.ii. On average, the average maximum daily temperature of inland towns drops by 0.011 of a degree for every metre above sea level the town is situated.

b. The correlation co-efficient of the regression line using latitude is significantly stronger than the equivalent co-efficient for the regression line using height above sea level.

$\therefore$ The equation using latitude is preferred.

12. Statistics, 2ADV S2 EQ-Bank 2

i. By calculator:

brain weight $= 49.4 + 2.68 \times$ body weight

COMMENT: Know this critical calculator skill!.

ii. Predicted brain weight of a dingo

$= 49.4 + 2.68 \times 22.3$

$= 109.164$

$= 109$ grams

13. Statistics, 2ADV S2 EQ-Bank 3

i. By calculator:

life span $= 42.89 - 2.85 \times$ sleeping time

COMMENT: Issues here? YouTube has short and excellent help videos - search your calculator model and topic - eg. "fx-82 regression line".

ii. Predicted life span of wallaby

$= 42.89 - 2.85 \times 4.5$

$= 30.06\ldots$

$= 30$ years

14. Statistics, 2ADV S2 EQ-Bank 1

a. Height

b. By calculator,

$$\text{Arm span } = 1.09 \times \text{height} - 15.63$$

COMMENT: Calculator skills for finding the least squares regression line were required in NESA sample exam - know this critical skill well!

c. On average, arm span increases by 1.09 cm for each 1 cm increase in height.

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15. Statistics, STD2 S4 EQ-Bank 4

i. By calculator,

$$\begin{aligned} r &= 0.94095\ldots \\ &= 0.941 \ \text{(to 3 d.p.)} \end{aligned}$$

COMMENT: Issues here? YouTube has short and excellent help videos - search your calculator model and topic - eg. "fx-82 correlation" .

ii. The association is positive and strong.

iii. x value $\Rightarrow$ foot length (independent variables)
y value $\Rightarrow$ height.
By calculator:
$$\text{Height} = 47.4 + 4.7 \times \text{foot length}$$

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16. Algebra, STD2 A2 2016 HSC 29e

i. 68 years

ii. Using (1900,60), (1980,80)

$$\begin{aligned} \text{Gradient} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{80 - 60}{1980 - 1900} \\ &= 0.25 \end{aligned}$$

♦♦ *Mean mark part (ii) 33%*

$\therefore$ After 1900, life expectancy increases by 0.25 years for each year later that someone is born.

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17. Statistics, STD2 S4 2013 HSC 28b

i.
$$\begin{aligned} \text{Gradient} &= \frac{\text{rise}}{\text{run}} \\ &= \frac{176 - 146}{16 - 11} \\ &= \frac{30}{5} \\ &= 6 \end{aligned}$$

ii. Males should grow 6cm per year between the ages 11-16.

♦♦ *Mean marks of 38%, 26% and 25% respectively for parts (i)-(iii).*
MARKER'S COMMENT: Interpreting gradients has been consistently examined in recent history and almost always poorly answered.

iii. Gradient $= 6$
Passes through $(11, 146)$

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ h - 146 &= 6(a - 11) \\ &= 6a - 66 \\ \therefore h &= 6a + 80 \end{aligned}$$

iv. If $a = 17$

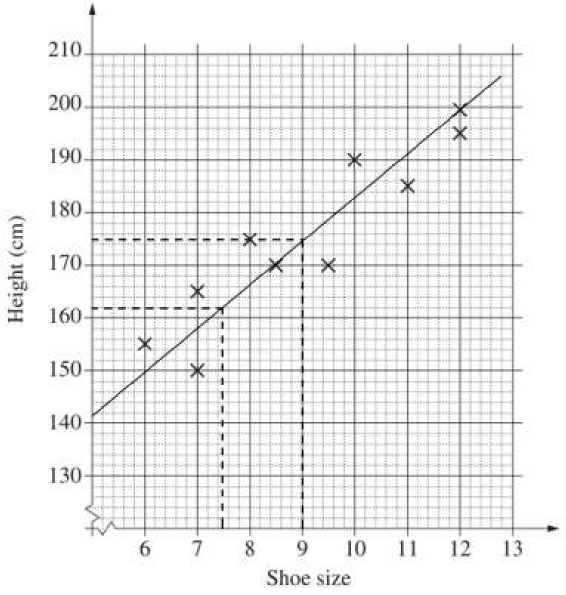
$$\begin{aligned} h &= (6 \times 17) + 80 \\ &= 182 \end{aligned}$$

$\therefore$ A typical 17 year old is expected to be 182cm

v. People slow and eventually stop growing after they become adults.

18. Statistics, STD2 S4 2015 HSC 28e

i.



ii. Shoe size $7\frac{1}{2}$ gives a height estimate of 162 cm (see graph)

Shoe size 9 gives a height estimate of 175 cm (see graph)

$$\therefore \text{ Height difference } = 175 - 162$$
$$= 13 \text{ cm}$$

iii. A correlation co-efficient of 1 would mean that all data points occur on the line of best fit which clearly isn't the case.

♦ *Mean mark 39%.*

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19. Statistics, STD2 S4 2019 HSC 23

a. Use "A + Bx" function (fx-82 calc):

$$r = 0.9811\ldots$$
$$= 0.98 \ \ (2 \text{ d.p.})$$

♦ *Mean mark 40%.*
COMMENT: Issues here? YouTube has short and excellent help videos - search your calculator model and topic - eg. "fx-82 correlation" .

b. Direction: positive

Strength: strong

c. $$\text{Height } = 0.866 \times 143 + 23.7$$
$$= 147.538 \text{ cm}$$

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20. Statistics, 2ADV S2 2020 HSC 27

$$y = -10.6063 + bx$$

Find b:

Line passes through $(\overline{x}, \overline{y})$

♦ *Mean mark 46%.*

$$\overline{x} = 22 - 0.525$$
$$= 21.475$$

$$\overline{y} = \frac{\text{total chirps}}{\text{number of data points}}$$
$$= \frac{684}{20}$$
$$= 34.2$$

$$34.2 = -10.6063 + b(21.475)$$
$$\therefore b = \frac{44.8063}{21.475}$$
$$\approx 2.0864$$

If $x = 19$,

$$y = -10.6063 + 2.0864 \times 19$$
$$= 29.03$$
$$= 29 \text{ chirps}$$

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21. Statistics, STD2 S4 2006 HSC 27b

i. The y-intercept occurs when $x = 0$. It has no meaning to have a height of 0 cm.

ii. A 20 cm height difference results in a foot length difference of 6 cm.

$\therefore$ A 10 cm height difference means George should have a 3 cm longer foot.

iii. A correlation co-efficient must be between –1 and 1. Foot length is positively correlated to a person's height and therefore isn't a negative value.

22. Statistics, STD2 S4 2014* HSC 30b

i. It indicates there is a strong positive correlation between the two variables

ii.
$$\begin{aligned}\text{IQR} &= Q_U - Q_L\\ &= 22.5 - 8.4\\ &= 14.1\end{aligned}$$

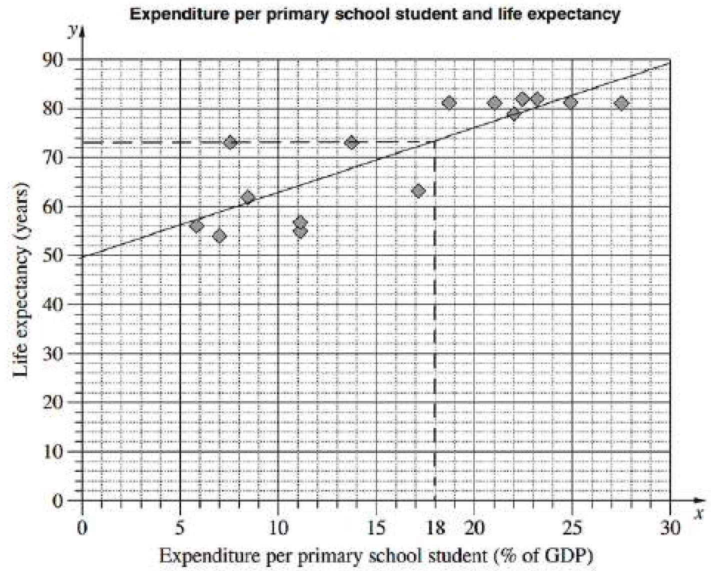
iii. An outlier on the upper side must be more than

♦ Mean mark 35%

$$\begin{aligned}&Q_u + 1.5 \times IQR\\ &= 22.5 + (1.5 \times 14.1)\\ &= 43.65\%\end{aligned}$$

$\therefore$ A country with an expenditure of 47.6% is an outlier.

iv.



v. Life expectancy ≈ 73.1 years (see dotted line)

♦♦ Mean mark 39%

Alternative Solution

When $x = 18$

$y = 1.29(18) + 49.9 = 73.12$ years

vi. At 60% GDP, the line predicts a life expectancy of 127.3. This line of best fit is only predictive in a lower range of GDP expenditure.

♦♦♦ *Mean mark 0%. The toughest question on the 2014 paper.*
COMMENT: Examiners regularly ask students to identify and comment on outliers where linear relationships break down.

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23. Statistics, STD2 S4 2009 HSC 28b

i. The correlation between height and mass is positive and strong.

♦ *Mean mark 48%.*

ii. Using $P_1(40, 1.2)$ and $P_2(80, 10.4)$

$$\begin{aligned}\text{Gradient} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{10.4 - 1.2}{80 - 40} \\ &= \frac{9.2}{40} \\ &= 0.23\end{aligned}$$

♦♦♦ *Mean mark 18%.*
MARKER'S COMMENT: Many students had difficulty due to the fact the horizontal axis started at $H = 40\text{cm}$ and not the origin.

Line passes through $P_1(40, 1.2)$

Using $y - y_1 = m(x - x_1)$

$$\begin{aligned}y - 1.2 &= 0.23(x - 40) \\ y - 1.2 &= 0.23x - 9.2 \\ y &= 0.23x - 8\end{aligned}$$

$\therefore$ Equation of the line is $M = 0.23H - 8$